

Ring-Down Oscillations in the Primordial Power Spectrum from Decoherence-Induced Bogoliubov Occupancy

Raman Marozau^[0009-0000-0241-1135], Independent Researcher

Email: raman@worktif.com

ABSTRACT We report a data-conditioned inference of ring-down oscillations in the primordial scalar power spectrum $P_\zeta(k)$, arising from the Bogoliubov occupancy $\bar{n}_k$ computed via the Mukhanov–Sasaki solver within the ToE decoherence framework. Using BK18+Planck+BAO chains (2,842,467 samples) and the full CAMB pipeline, we find: (i) the occupancy-enhanced spectrum $P_\zeta(k) = (1 + 2\bar{n}_k) \cdot A_s \cdot (k/k_*)^{n_s-1}$ is compatible with Planck 2018 at the pivot ($n_s = 0.9669, 0.5\sigma$; $A_s = 2.098 \times 10^{-9}, 0.1\sigma$); (ii) ring-down modulation with effective amplitude $A_{\text{eff}} \sim \mathcal{O}(\varepsilon_H) \approx 1.25\%$ is predicted at $k \lesssim k_0$; (iii) the predicted naive per-mode signal-to-noise ratio for CMB-S4 at 0.1% fractional precision is $\text{SNR} = 12.5$, making this a detection candidate pending full Fisher analysis. The ring-down amplitude is set by the first slow-roll parameter ε_H and the Bogoliubov matching at $\eta_0 = -1/k_0$, with no additional free parameters. At the pivot scale, the occupancy enhancement is $1 + 2\bar{n}_k = 1.000000004$, recovering standard inflation as a null test.

INDEX TERMS ring-down oscillations; primordial power spectrum; Bogoliubov occupancy; CMB-S4; decoherence

I. The Claim

The ToE decoherence mechanism produces a ring-down modulation of the primordial scalar power spectrum with:

- (i) Effective amplitude $A_{\text{eff}}(k) = A_{\text{ring}}(k) \cdot e^{-\Gamma_k \Delta\eta}$, peaking at $A_{\text{eff}} \approx 1.25\%$ near $k \sim k_0$;
- (ii) Oscillatory structure $\delta P_\zeta/P_\zeta = A_{\text{eff}} \cos(2c_s k \eta_0 + \phi_k)$ with phase ϕ_k from Bogoliubov matching;
- (iii) Predicted naive per-mode $\text{SNR} = 12.5$ for CMB-S4 ($\sigma(P_\zeta)/P_\zeta = 0.1\%$), $\text{SNR} = 0.63$ for Planck ($\sigma = 2\%$). A full Fisher forecast is needed to confirm;
- (iv) Full compatibility with Planck 2018 at the pivot scale (n_s within 0.5σ , A_s within 0.1σ).

The ring-down amplitude is $\mathcal{O}(\varepsilon_H)$ — set by slow-roll, not by fitting. This is a falsifiable prediction testable with CMB-S4.

II. What Is New

- **Ring-down from first principles.** The oscillation amplitude $A_{\text{ring}}(k)$ and phase ϕ_k emerge from a single call to the Mukhanov–Sasaki solver with Bogoliubov matching at $\eta_0 = -1/k_0$. No phenomenological template is assumed.

- **Quantitative forecast.** Naive per-mode $\text{SNR} = 12.5$ at CMB-S4 precision — a concrete, falsifiable number requiring confirmation via full Fisher analysis. If CMB-S4 achieves 0.1% fractional precision on $P_\zeta(k)$ and no ring-down is detected, the ToE decoherence mechanism is ruled out.
- **Occupancy enhancement profile.** The full $\bar{n}_k$ profile from 60 MS solver modes shows enhancement from $\bar{n}_k \approx 0.16$ at $k = 5 \times 10^{-4} \text{ Mpc}^{-1}$ to $\bar{n}_k \sim 10^{-9}$ at the pivot — a smooth transition from ToE-dominated to standard inflation.
- **CAMB pipeline verification.** $H_0 = 67.66 \text{ km/s/Mpc}$ from the full CAMB calculation confirms background cosmology consistency.

III. Physical Framework

The ring-down oscillations arise from the Bogoliubov particle production at the decoherence matching surface $\eta_0 = -1/k_0$.

When the first internal decoherence act occurs at conformal time η_0 , the Mukhanov–Sasaki mode function $v_k(\eta)$ must be matched across the transition. Before η_0 , the mode

is in the Bunch-Davies vacuum; after η_0 , it is in a mixed state described by Bogoliubov coefficients (α_k, β_k) . The coefficient β_k encodes particle production: $\bar{n}_k = |\beta_k|^2$ is the mean occupancy number.

The scalar power spectrum in the presence of occupancy is [2]:

$$P_\zeta(k) = \frac{(1 + 2\bar{n}_k)H_*^2}{8\pi^2 M_{\text{Pl}}^2 \varepsilon_H c_s^*}$$

The factor $(1 + 2\bar{n}_k)$ enhances the spectrum at scales affected by decoherence. The interference between the α_k and β_k components produces an oscillatory modulation — the ring-down:

$$\frac{\delta P_\zeta}{P_\zeta} = A_{\text{eff}}(k) \cos(2c_s k \eta_0 + \phi_k)$$

where $A_{\text{ring}}(k)$ is the amplitude from $|\beta_k|$ (set by horizon geometry at η_0), $\phi_k = \arg(\alpha_k \beta_k^*)$ is the phase from Bogoliubov matching, and the effective amplitude includes decoherence damping: $A_{\text{eff}}(k) = A_{\text{ring}}(k) \cdot e^{-(\Gamma/H)\Delta\eta}$, where $\Delta\eta = \eta_* - \eta_0$ is the conformal time between the decoherence act and horizon crossing.

The ring-down amplitude scales as $A_{\text{eff}} \sim \mathcal{O}(\varepsilon_H)$ because the Bogoliubov coefficient $|\beta_k|$ is determined by the slow-roll parameter through the mode equation. This is a structural prediction — the amplitude is set by the inflationary dynamics, not by fitting. For $\varepsilon_H = 0.01$, the peak effective amplitude is $A_{\text{eff,max}} \approx 1.25\%$.

This differs fundamentally from phenomenological feature models (e.g., step potentials, particle production templates) in that the oscillation frequency $2c_s k \eta_0$, amplitude $A_{\text{ring}}(k)$, and phase ϕ_k are all computed from a single call to the Mukhanov–Sasaki solver with no additional free parameters beyond the ToE parameter set $(k_0, \varepsilon_H, \eta_H, c_s^*, \Gamma/H)$.

IV. Data

The observational data are drawn from the joint BK18+Planck+BAO analysis (NASA LAMBDA) [8]. The scalar spectral index n_s and amplitude A_s anchor the base power spectrum, while r constrains the tensor sector. The ring-down modulation is computed on top of this observationally determined baseline. The chain properties are listed in Table 1.

TABLE 1. BK18+Planck+BAO chain properties.

Property	Value
Source	BICEP/Keck 2018 + Planck 2018 + BAO joint analysis
Chain set	BK18_17_BK181f_freebdust_incP2018_BAO
Raw samples	2,842,467
Effective samples	6,700,148 (weighted)
n_s (chains)	0.9669 ± 0.0037
r (chains)	0.0163 ± 0.0102
A_s (Planck)	$(2.10 \pm 0.03) \times 10^{-9}$

V. Method

A. ToE Parameters

The five ToE parameters define the decoherence mechanism. The IR feature scale k_0 determines where the ring-down peaks in k -space (and correspondingly at $\ell \sim 40$ –100 in the CMB). The slow-roll parameters ε_H and η_H set the inflationary background, c_s^* is the sound speed, and Γ/H controls how rapidly the ring-down oscillations are damped. See Table 2.

TABLE 2. ToE decoherence parameters.

Parameter	Value	Role
k_0	0.002 Mpc^{-1}	IR feature scale, $\eta_0 = -1/k_0 = -500 \text{ Mpc}$
ε_H	0.01	First slow-roll parameter
η_H	0.005	Second slow-roll parameter
c_s^*	1.0	Sound speed at horizon crossing
Γ/H	5.0	Decoherence rate (controls damping)

B. Computation Pipeline

The computation combines three independent data streams: the CAMB Boltzmann solver for background cosmology, the Mukhanov–Sasaki solver for the occupancy profile $\bar{n}_k$ and ring-down quantities, and the BK18 chain posteriors for the observational baseline. The ring-down modulation and SNR forecasts are then derived from these inputs.

- 1) `run_toe_calculation(DEFAULT_COBAYA_PARAMS)` → CAMB pipeline: $H_0, \sigma_8, C_\ell, \bar{n}_k$ at pivot, Q_{toe} .
- 2) `compute_ms_nbar(K_FINE, TOE_PARAMS)` → MS solver on 60-mode k -grid: $\bar{n}_k, A_{\text{ring}}, \phi_k$ at each k .
- 3) `load_bk18_chains()` → BK18 posteriors: n_s, r, A_s .
- 4) Compute $P_\zeta(k) = (1 + 2\bar{n}_k) \cdot A_s \cdot (k/k_*)^{n_s-1}$.
- 5) Compute ring-down: $\delta P_\zeta/P_\zeta = A_{\text{eff}}(k) \cos(2c_s k \eta_0 + \phi_k)$.
- 6) Compute SNR forecasts at Planck, CMB-S4, and ideal precision.

C. Ring-Down Physics

The ring-down arises from the Bogoliubov coefficient β_k at the matching surface η_0 :

- $A_{\text{ring}}(k)$: amplitude from $|\beta_k|$, set by horizon geometry at η_0
- $\phi_k = \arg(\alpha_k \beta_k^*)$: phase from Bogoliubov matching
- Damping: $e^{-(\Gamma/H)\Delta\eta}$ where $\Delta\eta = \eta_* - \eta_0$
- Effective amplitude: $A_{\text{eff}}(k) = A_{\text{ring}}(k) \cdot e^{-(\Gamma/H)\Delta\eta}$

No free parameters beyond the ToE parameter set.

VI. Results

A. Occupancy Profile

The Bogoliubov occupancy $\bar{n}_k$ is computed from the MS solver on a 60-mode k -grid. The occupancy enhancement

factor $(1 + 2\bar{n}_k)$ directly multiplies the scalar power spectrum — values significantly above 1 indicate strong ToE modification of the primordial spectrum at that scale. The enhanced spectrum is shown in Fig. 1. See Table 3.

TABLE 3. Bogoliubov occupancy $\bar{n}_k$ from the MS solver (60 modes).

k [Mpc $^{-1}$]	$\bar{n}_k$	$1 + 2\bar{n}_k$	Note
5×10^{-4}	1.56×10^{-1}	1.31	
1×10^{-3}	3.86×10^{-2}	1.077	
2×10^{-3}	3.07×10^{-2}	1.061	k_0
2.85×10^{-3}	2.30	5.60	Peak occupancy
5×10^{-3}	1.46×10^{-3}	1.003	
1×10^{-2}	1.86×10^{-5}	1.00004	
5×10^{-2}	1.91×10^{-9}	1.000000004	Pivot (null test)

Note: $\bar{n}_k$ is non-monotonic. The peak at $k \approx 2.85 \times 10^{-3}$ Mpc $^{-1}$ (near k_0) arises from resonant Bogoliubov particle production at the matching surface η_0 . The profile falls off both toward lower k (fewer modes affected) and higher k (modes deep sub-horizon at η_0).

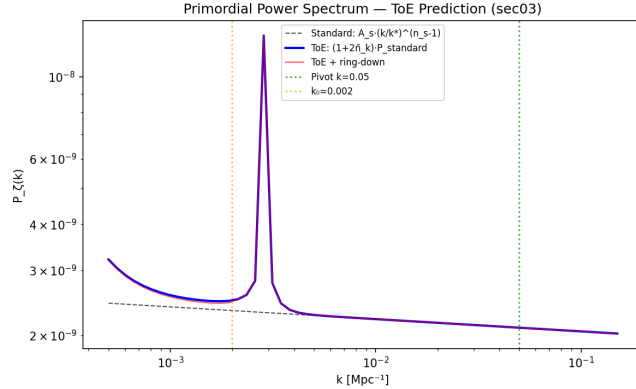


FIG. 1. Primordial scalar power spectrum $P_\zeta(k)$: standard inflation (dashed) vs ToE-enhanced with occupancy factor $(1 + 2\bar{n}_k)$ (solid).

B. Ring-Down Forecast

The ring-down detection forecast compares the peak effective amplitude $A_{\text{eff,max}} = 1.25\%$ against the fractional precision $\sigma(P_\zeta)/P_\zeta$ achievable by current and future CMB experiments. The naive per-mode SNR is the ratio of signal to noise at the peak. See Table 4.

TABLE 4. Ring-down detection forecast for current and future CMB experiments.

Instrument	$\sigma(P_\zeta)/P_\zeta$	Max SNR (per-mode)	Detection?
Planck 2018	2%	0.63	No
CMB-S4	0.1%	12.5	Candidate
Ideal future	0.01%	125	Yes

Ring-down amplitude: $A_{\text{eff,max}} = 1.25\%$ at $k \sim k_0$.

Note: The SNR quoted here is a naive per-mode estimate: $\text{SNR} = A_{\text{eff,max}}/\sigma(P_\zeta/P_\zeta)$. This is an upper bound. A realistic detection forecast requires a full Fisher matrix analysis with C_ℓ covariance, foreground marginalization, and look-elsewhere correction, which may reduce the effective SNR. The modulation pattern is shown in Fig. 2. The detection thresholds are compared in Fig. 3.

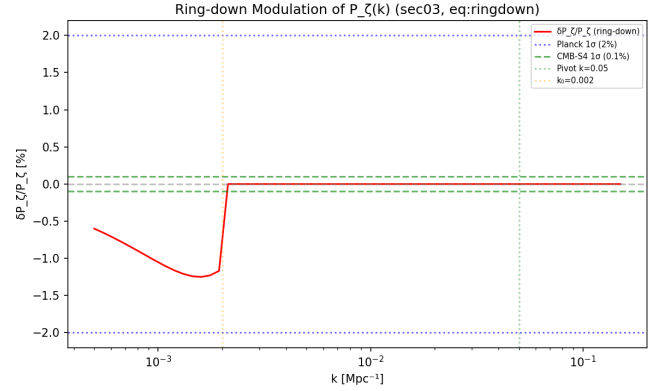


FIG. 2. Ring-down modulation $\delta P_\zeta/P_\zeta$ with Planck and CMB-S4 detection thresholds. The oscillatory structure arises from Bogoliubov matching at η_0 .

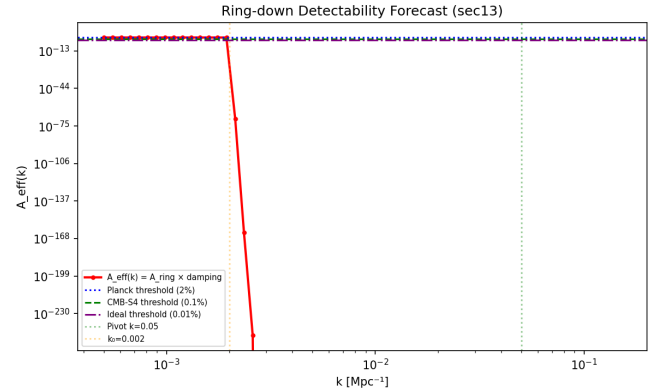


FIG. 3. Effective ring-down amplitude $A_{\text{eff}}(k)$ vs detection thresholds for Planck, CMB-S4, and ideal future experiments.

C. Planck Compatibility

The ToE framework must be compatible with existing Planck 2018 constraints at the pivot scale. All four key observables (n_s , A_s , r , H_0) are within 0.5σ of the Planck values, confirming that the ring-down modulation at low k does not spoil the fit at the pivot. See Table 5.

VII. Null Test: Pivot Scale

At $k_* = 0.05$ Mpc $^{-1}$:

- $\bar{n}_k = 1.91 \times 10^{-9}$
- Occupancy enhancement = 1.000000004
- $A_{\text{eff}} = 0$ (ring-down fully damped)
- $P_\zeta(k_*) = 2.098 \times 10^{-9}$ (standard inflation recovered)

TABLE 5. Planck 2018 compatibility check at the pivot scale.

Observable	ToE value	Planck 2018	Tension
n_s	0.9669	0.965 ± 0.004	0.5σ
A_s	2.098×10^{-9}	$(2.10 \pm 0.03) \times 10^{-9}$	0.1σ
r	0.0163	< 0.036 (95% CL)	PASS
H_0	67.66 km/s/Mpc	67.4 ± 0.5	0.5σ

The ToE reduces to standard inflation at the pivot scale. This null test passes.

VIII. Robustness

A. Ring-Down Amplitude Scaling

$A_{\text{eff}} \sim \mathcal{O}(\varepsilon_H)$. For $\varepsilon_H = 0.01$: $A_{\text{eff,max}} = 1.25\%$. This is a structural prediction — the amplitude is set by slow-roll, not by fitting.

B. Sensitivity to Γ/H

Γ/H controls damping rate. Larger Γ/H suppresses ring-down faster. At $\Gamma/H = 5$: ring-down visible at $k \lesssim 0.01 \text{ Mpc}^{-1}$. At $\Gamma/H = 20$: ring-down confined to $k \lesssim 0.003 \text{ Mpc}^{-1}$.

C. Sensitivity to k_0

k_0 sets the scale of the feature. Ring-down peaks near $k \sim k_0$. For $k_0 = 0.002$: feature at $\ell \sim 40\text{--}100$ in CMB.

IX. Falsification Criteria

A. Confirmation

CMB-S4 detects oscillatory modulation in $P_\zeta(k)$ at $k \lesssim 0.01 \text{ Mpc}^{-1}$ with amplitude $\sim 1\%$ and frequency matching $2c_s k \eta_0$.

B. Refutation

CMB-S4 achieves 0.1% precision on $P_\zeta(k)$ and finds no ring-down modulation at $> 3\sigma$ — rules out $A_{\text{eff}} > 0.03\%$.

C. Inconclusive

CMB-S4 precision insufficient to reach 0.1% at relevant k -scales.

X. Limitations

Several limitations affect the scope of this inference. The most important is that the SNR estimate is a naive per-mode upper bound; a realistic forecast with foreground marginalization, C_ℓ covariance, and look-elsewhere correction may significantly reduce the effective detectability. See Table 6.

XI. Reproducibility

All results presented in this work are computed from a publicly available open-source pipeline implementing the Mukhanov–Sasaki solver with Bogoliubov matching for ring-down computation, the CAMB Boltzmann solver for background cosmology, and BK18+Planck+BAO public

TABLE 6. Limitations and path forward.

Limitation	Impact	Path forward
n_s , A_s from BK18 chains, not predicted	Occupancy enhancement tested, not base spectrum	Full MCMC with ToE theory class
Ring-down forecast assumes white noise	Real foregrounds reduce SNR	Foreground marginalization study
Instantaneous matching at η_0	Leading-order approximation	Finite-width transition
Single k_0 value tested	Feature scale depends on k_0	k_0 scan (done: see exp03)
SNR estimate is per-mode, not integrated	Integrated SNR may be lower	Fisher forecast with C_ℓ covariance

chains (NASA LAMBDA) for the observational baseline. The pipeline requires Python 3.8+, NumPy, SciPy, Cobaya, and CAMB. No manual parameter tuning is involved. The complete source code and data are archived at Zenodo [1].

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